Introduction to Stack

Stack is a linear data structure that follows a particular order of operation. This order can either be **Last In First Out** (**LIFO**) or **First In Last Out** (**FILO**).

Imagine you have a pile of books that you plan to read. You keep adding books to the top of the pile. When you're ready to start reading, you take a book from the top of the pile. The last book you added to the pile is the first one you read. That's LIFO - the principle that stack data structures operate on.

What makes stacks so unique is their simplicity and elegance. Despite being straightforward, they can be incredibly powerful when used in the right way.

Real-world Examples of Stacks

Before diving into technicalities, let's familiarize ourselves with stacks in our daily lives. Stacks are everywhere around us, even if we might not notice them. Here are some examples to help you relate:

1. **Stack of Books**: This is perhaps the simplest example. A pile of books is a stack. The book on the top was the last one added and will be the first one removed.
2. **Stack of Plates**: Picture a stack of plates at a buffet. The first plate you'll take is the one on top, which was the last one put on the stack.
3. **Web Browser History**: Every time you visit a new webpage, it's added to the top of your history stack. When you hit the back button, you're "popping" pages off the top of the stack.
4. **Undo Function in Software Applications**: The undo function in software applications uses a stack to remember actions. The most recent action is on top and will be the first one undone.

Looking at these examples, it's clear that stacks are not just a theoretical concept, but a practical one that we use unconsciously in our daily lives. With this understanding, let's dive deeper into the operations that define stacks in data structures.

Stack of books

The LIFO Principle

As mentioned earlier, stacks in data structures operate on the Last-In, First-Out (LIFO) principle. This means that the last item added to the stack is the first one that gets taken out.

The LIFO principle is the heart of stack data structures. It governs how we add and remove elements, making stack operations predictable and consistent. With that, let's explore the primary operations that you can perform on a stack.

There are four key operations that you can perform on a stack:

1. **Push**: This is how we add an element to the stack. The element is always added to the top.
2. **Pop**: This is the removal of an element from the stack. The element removed is always from the top.
3. **Peek or Top**: This operation allows us to see the element on the top of the stack without removing it.
4. **IsEmpty**: This operation checks if the stack is empty.

Let's see these stack operations in detail in the next section.

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